

Advanced (Carolina Reaper)

Q1. Solve the system: $x + y = 4$, $x - y = 2$

- (A) $x = 3, y = 1$
- (B) $x = 2, y = 2$
- (C) $x = 1, y = 3$
- (D) $x = 4, y = 0$
- (E) $x = 0, y = 4$

Q2. Solve the system: $2x + 3y = 7$, $x - 2y = -3$ using elimination

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) No solution

Q3. Solve the system: $x + 2y = 5$, $3x - 2y = 7$

- (A) $x = 3, y = 1$
- (B) $x = 2, y = 2$
- (C) $x = 1, y = 3$
- (D) $x = 4, y = 0$
- (E) $x = 0, y = 5$

Q4. The system $x + y = 2$, $2x + 2y = 4$ has

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q5. Solve the system: $x - y = 1$, $2x + 2y = 0$

- (A) $x = 1, y = 0$
- (B) $x = 0, y = 1$
- (C) $x = 2, y = 1$
- (D) $x = 1, y = -1$
- (E) No solution

Q6. The system $2x + 3y = 7$, $4x + 6y = 10$ has

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q7. Solve the system: $x + 2y = 4$, $2x - 2y = 2$ using substitution

- (A) $x = 2, y = 1$
- (B) $x = 1, y = 2$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) No solution

Q8. The system $3x - 2y = 5$, $6x - 4y = 8$ has

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Open Ended Questions (FRQ)**Q9.** Solve the system: $x + 3y = 5$, $2x - 2y = 4$. Show all work and explain your reasoning.**Q10.** Consider the system: $2x + y = 3$, $x - 2y = -3$. Determine the number of solutions and explain why.

Chef's Tips

- Check for correct units and significant figures
- Verify calculations and algebraic manipulations
- Ensure explanations are clear and concise